NaReT: what Denis built and what changed

A practical comparison of the original app and the merged headset app.

Includes scenario rules, speech, CCBM, tracking, CPR feedback, patient appearance and menus.

Original Marek dialogue: all 14 fixed lines, in German and Polish, with English translations.

Credit the existing work correctly

Denis built the Godot/OpenXR application, scenario engine, patient and witnesses, audio/cloud framework, menus and feedback infrastructure.

His setup already imported your hand-zone, compression detector and pose stabilizer. He also integrated an ArUco addon and settling placement.

The merge adds your complete three-marker/startup/motion-gate path, restores your CPR display, connects Paul’s detector, and adds English and CCBM.

Same state machine; a different way to submit actions

Before: progression buttons submitted actions to Denis's scenario controller.

Now: speech → recognised text → observation → CCBM decision → Denis's allowed-action guard → scenario update.

The state machine still decides what is legal. The language model cannot freely move the exercise to any stage.

Exactly what the two switches do

cpr_witness_only: exclude AED actions and treatment performed by a virtual bystander. Anna stays; Marek and the AED scene instances are removed.

observation_driven: hide routine progression buttons, such as responsiveness, breathing, ventilation and handover.

Start, reset, calibration and audio controls remain. Physical compression events still come from your detector.

Marek: what his original role was

Denis’s original second witness could call 112, fetch an AED, relay dispatcher advice and offer/help with compressions.

His 14 fixed lines were triggered by events and timers; they were not a conversation played from start to finish.

The current one-witness exercise does not instantiate him. His original source remains in the repository. The full script is included later.

Local recognition: what was actually added

Quest microphone → Denis's existing WAV capture → uni_api_client.gd local ASR route.

USB forwarding sends localhost:8767/inference to the PC's whisper.cpp server in Ubuntu/WSL. It runs the medium model on the PC.

Text returns to the app. Cloud LLM/TTS support remains separate. Local ASR is not bundled inside the APK.

Recognition and interpretation are separate

ASR answers: “What words were spoken?”

The observation mapper/interpreter answers: “Was this a completed finding, a request, a question, or just conversation?”

CCBM evaluates the observation history; Denis’s scenario guard checks the resulting action. Saying “I gave a breath” remains a reported action, not a sensor measurement.

Denis did more than ArUco detection

Original: an ArUco adapter, configurable marker, tracking status, temporal filter and model calibration. His spatial anchor settled for 1.5 seconds and averaged a stable window.

Added: calibrated three-marker common-anchor fusion, robust startup median/rotation medoid, viewer-motion gating and restored original filter settings.

Paul: newer native detector/addon. Your camera adapter and downstream placement remain in the merged path.

The complete current placement path

Camera colour conversion → Paul's detector → capture-time room pose

→ marker_layout.gd: each marker reconstructs the same anchor

→ startup_placement.gd: robust initial reference

→ viewer_motion_gate.gd → original simple_pose_stabilizer.gd

→ spatial_anchor.gd: saved alignment → Denis's patient model

Transparency is implemented in a shader

mannequin_controller.gd installs patient_overlay.gdshader on the patient materials.

A 4 × 4 pixel pattern discards a fraction of body pixels. Remaining pixels keep normal opaque depth; this avoids showing multiple transparent layers through each other.

The head region stays solid to avoid teeth/eyeballs showing through facial skin. The UI converts transparency to opacity: opacity = 1 − transparency / 100.

Alignment, floor lock and fitting are three different things

Alignment: patient world pose = marker anchor × saved model offset. Controller nudges edit the relative position/rotation.

Floor orientation: discard marker pitch/roll. Floor height/contact: separate reference-height and posed-model geometry handling. One does not guarantee the other.

Optional fitting: target head-to-waist length and shoulder width can scale the model. Defaults are zero (disabled); scene scale 0.9 does not prove a measured torso match.

Head animation: the current values differ from Denis’s

Denis: neck up to 28°; head up to 20°. Current merged code: 10° + 10°. The reduction was a prototype choice, not evidence that Denis’s animation was wrong.

Recommendation: restore 28°/20° after checking the rig’s direction, while keeping the deliberate two-hand gesture trigger.

These are local bone rotations. They do not prove the physical mannequin’s airway was opened.

Menu audit: where the repetition occurs

The old tiny language and microphone controls are hidden. The large language/microphone controls are active.

Start can still appear both as the large Start control and in the dynamic action list. Training controls also repeat in the calibration panel.

Choosing a language currently also requests Own-language mode; this coupling makes the behavior harder to understand.

One Language umbrella; one training control

Start training / Training active

Language: English → English • Deutsch • Polski

Microphone: On / Off

Align patient → position, rotation, Save

More → Test voice, training mode, results/export

Keep the chosen option green. Flash momentary actions briefly. Use the same selection state in the controller-accessible menu.

Executive assessment

Denis supplied the application architecture, patient/witness presentation, scenario engine, audio/cloud framework, interface, and feedback infrastructure. His dependency setup already imported the user's compression/hand-zone code and simple_pose_stabilizer.gd. Those were not invented during this merge. He also already integrated an external ArUco addon.

Executive assessment

The merge changes the headset placement pipeline, connects the original CPR display, narrows the exercise to one witness without AED, adds English, and routes reported exercise actions through a new CCBM bridge. Paul's imported addon supplies the newer detector. The latest fix activates the user's tuned stabilizer settings and rest-pose policy, which had been missing in the adapter.

Executive assessment

Source inspection establishes implementation, not correct behavior on hardware. All 24 application test suites passed in the latest run, with the Windows-native detector excluded. The imported Windows DLL is blocked by Application Control. Android detection was observed previously with all three colored markers. The latest restored stabilizer still needs the user's stationary/head-motion check; this review does not claim the entire exercise is finished.

1. Composition and ownership — scripts/main.gd

Before: constructed Denis's scenario, QCPR manager, EKG, AED, two-witness director, cloud/audio modules, tracker, interaction, room layout and UI.

1. Composition and ownership — scripts/main.gd

Now: sets cpr_witness_only and observation_driven; constructs the CCBM bridge and observation interpreter; instantiates a single-witness director with Anna and null second rescuer/AED references. EKG and patient systems remain. The current startup does not instantiate QCPR manager or the AED object.

1. Composition and ownership — scripts/main.gd

The tracker receives the marker-layout profile. On the headset, the spatial anchor receives the viewer head and the tracker-level temporal filter is disabled so the anchor owns stabilization. This avoids two filter stages.

1. Composition and ownership — scripts/main.gd

Audio recording and playback mute the CPR metronome. English confirmations, recognized-text display, service status messages, controller nudging, save actions, witness positioning and accessible training controls are connected here. Virtual room/floor and colored hand meshes are not installed in the XR flow.

1. Composition and ownership — scripts/main.gd

Review: this remains the largest integration file. AED/QCPR declarations and legacy branches remain even though those systems are not instantiated. They are inactive leftovers, not evidence that real QCPR measurements are connected. Start requires placement and saved calibration. Exact headset use should be tested in one complete run rather than inferred from individual functions.

2. Scenario — scripts/scenario/scenario_controller.gd

Before: deterministic assessment/call/CPR/AED/recovery states, counts and allowed-action guards. Buttons could submit training actions.

2. Scenario — scripts/scenario/scenario_controller.gd

Now: adds two policy switches. Witness-only mode rejects AED actions and bystander-generated treatment events. Observation-driven mode hides routine clinical progression buttons through get_interface_actions() while retaining setup/reset controls. The original scenario guard still validates actions after CCBM acceptance. Logging accepts CCBM probability and spoken-report metadata. The summary identifies the focused exercise as cpr_single_witness_practice.

2. Scenario — scripts/scenario/scenario_controller.gd

Review: AED enums and implementation remain for the broader scenario; they are excluded in the selected mode. This is a focused CPR exercise, not Denis's full original AED simulation. Speech confirms reported findings, not physical technique. Physical compressions remain detector events.

scripts/scenario/bystander_director.gd — unchanged upstream implementation

Denis's original two-witness director remains in the repository. The current app instantiates a subclass instead, so presence of this file does not mean Marek or automated bystander compressions are active.

scripts/scenario/single_witness_director.gd — new

Overrides state/event handling: Anna arrives, urges action after waiting and responds to accepted responsiveness assessment. Stroke and intent callbacks do nothing. It never supplies compressions or fetches an AED. It deliberately reuses the original director's timing/audio infrastructure.

scripts/ai/llm_orchestrator.gd — modified

Adds English fallback replies, English model prompting/response validation and a single_witness policy. In that mode, AED-fetch replies are suppressed and the prompt forbids another helper or witness-performed CPR. Its queue and dialogue-only separation are retained. Generated replies do not advance the scenario. Calm/thank replies update Anna's emotion, not proof of a learner action.

scripts/ai/llm_orchestrator.gd — modified

Review: a cloud reply and a prerecorded fallback are different paths. Arbitrary conversation requires the configured LLM; local ASR alone does not provide open-ended dialogue. Legacy AED response dictionaries remain but are filtered in the active mode.

scripts/ai/audio_manager.gd — modified

English is accepted throughout recording/playback. An empty capture buffer now returns zero level instead of retaining a stale loud value; this prevents a silence timer from being continually reset by old samples. Recording does not restart an already-playing microphone stream. An unscripted missing clip does not substitute an unrelated Anna calming line. Original microphone capture, voice activation, playback queue and prepared-clip support remain.

scripts/api/uni_api_client.gd — modified

Adds optional local ASR at http://127.0.0.1:8767/inference, a separate request with a 30-second timeout, no cloud credentials, no redirects and English transcription support. Denis's cloud completion/TTS/authentication framework remains. The local route is a USB-forwarded development deployment, not an independent speech recognizer running inside the Quest.

scripts/scenario/spoken_observation.gd — new

Normalizes speech and maps bounded statements/requests to structural action codes. Supports selected multi-sentence forms around a finding; does not mean every wording or pace is accepted. Speech cannot create a compression stroke.

scripts/scenario/speech_confirmation.gd — new

English-only pending proposals last 30 seconds and are bound to the current scenario context. Yes/no applies only to an explicit pending proposal. A small list handles observed transcription mistakes and tentative statements. It is not a general fuzzy-match acceptance engine.

scripts/ai/observation_interpreter.gd — new

For speech not handled by bounded matching, asks the configured LLM for an action, kind, result and verbatim supporting evidence. Validates the schema, evidence substring and expected result. Questions, intentions, ambiguous or contradictory statements cannot silently become completed treatment actions. Conversation is sent onward to Anna's dialogue system.

scripts/ai/observation_interpreter.gd — new

Review findings: the interpreter has one in-flight request; speech arriving while busy gets clarification, not a queued interpretation. Without a configured LLM, unfamiliar wording receives clarification. English confirmation behavior is not mirrored completely in German/Polish. Recognition at different speaking rates and interruption during playback still needs recorded headset validation.

scripts/scenario/ccbm_bridge.gd

Sends structural observation history to an external worker. Requires a response identifying engine ccbm, the requested action and a finite probability at least 0.90. Then checks the action is still allowed by Denis's scenario. One request is allowed at a time; history is limited to 256 observations. Reset invalidates old responses. It records compression onset from accepted tracked strokes rather than sending every stroke over HTTP. Reported ventilation is still a spoken claim. Start does not require CCBM inference.

integrations/ccbm/d.pddl, p.pddl, d-observations.cpp, server.py

Added behavior model/problem/observation code and HTTP wrapper around the compiled CCBM worker on the PC/Linux runtime. The bridge is not itself the inference engine. Repeated breathing observations represent first/second breath in the model. Local speech/CCBM forwarding is configured by development tools.

integrations/ccbm/d.pddl, p.pddl, d-observations.cpp, server.py

Review: 0.90 is explicitly a prototype policy, not a threshold selected by data-driven validation. Endpoint/model availability and accepted exercise transitions must be checked during a full run. CCBM cannot prove a physical breath, correct hand position or correct airway maneuver from a spoken report.

scripts/tracking/aruco_tracker.gd — modified Denis adapter

Adds profile loading and marker combination, used-marker evidence reporting and a switch for temporal filtering. Preserves marker size/ID configuration, tracking status and desktop preview. Chooses the migrated camera provider in the XR path. Combining markers uses the exact newest detection result, not unrelated stale poses. Unrecognized IDs are ignored by the layout.

scripts/tracking/user_camera_provider.gd — new adapter from original approach

Retains CameraServer capture, background detection and capture-time head-pose world conversion. Uses the imported detector's detect_markers() when present, explicitly selects 4x4/50, passes configured physical marker sizes and retains the existing intrinsics. Identity head pose is passed to detection; the saved head transform is applied once. Old API fallback remains.

scripts/tracking/user_camera_provider.gd — new adapter from original approach

The explicit lens compatibility correction logged as identity on the headset: the suspected inverse difference was ruled out. Diagnostic PNG saving is off. Detection timing is periodically logged. Camera latency remains a fixed 50 ms estimate; the addon's accurate CameraX timestamp path is not integrated here.

scripts/tracking/colored_camera.gdshader — new

Combines luminance/chroma camera textures and classifies blue to white and orange to black before detection. Full-resolution processing remains enabled for colored markers. Lighting, printing and color thresholds can still affect detection; this is not proof that all resolutions were optimized.

scripts/tracking/colored_marker_image.gd — new

CPU fallback for non-GPU images, with RGB hue classification and luminance fallback handling. Unit tests cover mapping and image handling; these are not a hardware lighting benchmark.

scripts/tracking/marker_layout.gd — added during migration

Loads calibrated marker-to-common transforms and a common-to-sternum bridge. Each visible marker reconstructs the same anchor; translations are averaged and sign-aligned quaternions combined. Reports position spread. Optional floor orientation removes pitch/roll; it does not place the mesh bottom on the floor.

scripts/tracking/startup_placement.gd — added during migration

Robust coordinate-median position and measured rotation medoid. Starts checking after 30 independent readings, at three-reading checkpoints, with 1.5 mm / 1-degree estimate stability; four stable comparisons permit earliest acceptance at 42 readings. A 100-reading fallback exists. Faster detection changes the wall-clock duration of observation-count rules.

scripts/tracking/viewer_motion_gate.gd — migrated original gate

Holds during head movement using a 0.3 s head window, 0.15 m/s and 20 degrees/s, then 0.5 s settling. Five readings learn viewpoint bias within 10 cm AND 10 degrees. Small updates are held beyond 1.8 m; a 30 cm OR 25-degree change with three full-evidence detections has an override. These heuristic constants were retained, not newly proved optimal by this review. The experimental blanket partial-marker hold was removed at the user's request.

scripts/tracking/spatial_anchor.gd — modified Denis placement

Denis's original one-marker 1.5 s settling/mean path remains for desktop. The headset uses the robust startup and viewer gate. Model offset and save/load compatibility remain. Adds floor reference/geometry handling, model orientation, opacity persistence, height preservation and explicit reposition resets.

scripts/tracking/spatial_anchor.gd — modified Denis placement

Version 42 finally configures the original filter: window 5, position dead zone 20 mm, rotation dead zone 0.3 degrees, smoothing 1.2 s, prior 2 s; stable endpoint checks 1.5 mm / 1 degree over 3 observations and relocation floors 80 mm / 8 degrees. A separate remembered rest pose is used and stable components can re-anchor independently. Earlier statements that the script's presence meant the full original policy was active were incomplete.

scripts/tracking/marker_pose_gizmo.gd — new diagnostic

White marker-center dot, labeled ID and 5 cm RGB axes on raw world marker poses, before patient/layout offsets. Hides after 500 ms without detection. Depth-test bypass keeps it readable through the patient. Raw gizmo jitter is intentional.

scripts/tracking/openxr_setup.gd / room_layout.gd — modified

Adds floor-height helpers respecting active tracking spaces, reference transforms and live tracked room-anchor positions. Original XR/scene integration remains. room_anchor.gd itself is unchanged.

scripts/tracking/openxr_setup.gd / room_layout.gd — modified

Review: three issues should not be conflated: marker pose alignment, mesh alignment offset, and stationary smoothing. A floor orientation constraint is also different from physical mesh-floor contact. Current gate heuristics and 50 ms camera timing can still cause viewpoint-dependent shifts. Real movement, occlusion and rest reacquisition need hardware checks without inventing offsets.

scripts/tracking/xr_interaction.gd — modified

Denis already instantiated the upstream hand-zone/stroke detector. Changes now attach the original CPR HUD to the XR camera; use its BPM interval source rather than a separate adapter calculation; retain the original zone dimensions; enable metronome except while audio is busy; reset the 0–30 practice count for a new cycle. The standalone automatic breathing-pause scheduler stays disabled because the scenario owns the breathing events.

scripts/tracking/xr_interaction.gd — modified

The practice display/zone appears after patient placement, but only CPR-stage strokes receive exercise credit. Adds controller trigger menu clicks while retaining hand pinch/poke input. Controllers do not generate physical CPR counts.

addons/naret_upstream/avatar/*

cpr_hand_zone.gd/.tscn, cpr_motion_session.gd, cpr_metronome.gd and the target guide originated in the user's project and were already dependencies of Denis's application. cpr_feedback_hud.gd was brought into this runtime for the original 0–30 count, pace needle, depth bar and coaching. The stabilizer script matches the original project; the important latest change is its configuration and rest-pose use in the spatial anchor.

addons/naret_upstream/avatar/*

Review: depth is hand excursion, not measured mannequin compression depth. Speech cannot add compression counts. The existing sensor modules would need real hardware/plugin connection for verified depth/recoil/ventilation data.

scripts/mannequin/mannequin_controller.gd — modified

Adds optional anatomical fitting, resized mouth/forehead targets and neutral gesture landmarks. Fit dimensions default to zero; scene model scale is 0.9. Therefore the presence of fitting code does not establish a measured mannequin match. Preserve the actual physical torso dimensions when assessing fit.

scripts/mannequin/mannequin_controller.gd — modified

Adds body overlay materials using assets/patient_overlay.gdshader; head region stays solid to avoid seeing teeth/eyeballs through facial skin. Body opacity uses a 4x4 screen-space coverage mask instead of layered alpha blending. Pallor updates preserve material alpha. A pattern can remain visible, especially in XR.

scripts/mannequin/mannequin_controller.gd — modified

Important exact difference: current head-tilt limits are 10 degrees neck and 10 degrees head. Denis's baseline uses 28 and 20. These values are not the same, even though earlier conversation discussed retaining Denis's angles.

scripts/tracking/head_tilt_gesture.gd — new

Two separate tracked hands: forehead palm within 6 cm, other index tip within 4.5 cm of chin; 0.35 s positioning; coordinated movement within 2 s, sustained 0.12 s; 0.25 s lost-tracking grace. Limits are provisional. It triggers animation only and does not score or confirm actual airway opening. Replaces Denis's one-palm-within-10-cm trigger.

Other mannequin scripts — unchanged

AED device, breathing system, bystander controller, heart system, pupil system, ragdoll and rig animator retain upstream implementations. Whether each is active depends on composition. AED code remains inactive in the focused exercise; retained physiological displays are simulations, not sensed patient physiology.

scripts/ai/language_manager.gd — modified

Adds English scenario/action lines and UI translation mapping, three-language selection and some English communication intents. Denis's DE/PL content remains. Missing UI translation keys fall back to German, so full English coverage is not guaranteed merely by selecting EN.

scripts/ui/hand_menu.gd — modified

Larger direct training, language, microphone and voice-test controls alongside the wrist-menu infrastructure. Uses observation-driven action lists. Language and active-state feedback turn green; momentary actions flash rather than remaining selected. Some visible strings remain English regardless of language.

scripts/ui/calibration_panel.gd — modified

Retains XYZ and Euler rotation preview through the model offset, preserves scale, and gates saving on startup placement. Removed marker geometry, scan visibility and model-scale controls. Later requests added language/start/audio/microphone controls outside scrolling, plus body transparency and floor adjustment fields. It is therefore broader than the earlier requested minimal six-field panel. Close keeps preview; Save writes it. Y controls can be disabled when floor locked. Hard-coded English messages need localization review.

scripts/ui/info_panel.gd — modified

Adds next-step/example phrases and hand-compression display. Helps the learner understand expected observations instead of requiring hidden action buttons. Remaining literal English strings require DE/PL checks.

scripts/ui/dashboard.gd — modified

Adds EN selection and localized labels, hides QCPR simulator button, uses observation-driven available UI actions and labels hand depth as an estimate. Desktop/simulation utilities remain; their presence is not headset sensor support.

scripts/ui/briefing_panel.gd — modified

Skips missing-AED feedback in witness-only mode, adapts the final advice for that scenario and routes localization through the language helper. This file changes what is displayed; it does not replace the scoring or session-log engine.

10. Feedback, EKG, QCPR and utilities retained from Denis

score_tracker.gd and session_logger.gd are unchanged upstream. The input events and scenario metadata are extended, so their output can differ even without editing the files. Full results should be checked for a no-AED, speech-report session. Numeric verified-quality scores must not imply measured chest depth.

10. Feedback, EKG, QCPR and utilities retained from Denis

ekg_system.gd and ekg_renderer.gd are unchanged. performance_monitor.gd is unchanged. The QCPR backend, real BLE adapter, constants, packet parser, manager, simulated backend and QCPR debug panel are unchanged source; current main does not start their runtime. Native BLE availability is a separate integration task.

Scene, addon and build changes outside scripts/

- scenes/main.tscn: selects data/user_marker_layout.cfg, supplies the default 180-degree X model orientation, and removes the AED and Marek scene instances. This is an actual scene change, not just hiding their buttons. - scenes/mannequin.tscn: sets the human model scale to 0.9. Saved model offsets can add their own transform, so the scene scale alone is not a torso measurement. - assets/patient_overlay.gdshader: new opaque-depth coverage shader with a solid head region; discussed in section 8. - addons/opencv_aruco: imported from Paul's addon-port branch, commit 729f8c0, with new packaged native libraries/API and export guard. It is a selective addon port, not replacement of Denis's whole project. Its standalone timestamped CameraX tracker is not the active camera provider. - project.godot: the OpenCV editor-plugin section was moved; comparison with this baseline does

Scene, addon and build changes outside scripts/

not show a new XR-reference-space policy in this file. - export_presets.cfg: installed build identifies as code 42, 0.3.40-original-stabilizer; application package remains the same so replace installation preserves user data. - English script/audio and service setup tools are additions. Service settings are runtime/device configuration, not evidence that all required servers are bundled in the APK. No credential values are included in this review.

Validation still required

1. Confirm the latest patient stabilization on a stationary mannequin, head movement, partial marker loss and restart. Compare patient versus raw gizmos. 2. Confirm saved model fit/nudge and floor-height behavior after restarting and XR recentering; do not compensate a camera-pose defect with a model offset. 3. Decide whether to retain the current 10+10-degree visual animation or restore Denis's 28+20-degree animation; test its direction on the actual rig. 4. Run one full English exercise with natural speech, interruptions, two breaths, compression cycle, handover and debrief. Log ASR, CCBM and scenario decisions. 5. Repeat in German/Polish; review translated UI, audio availability and fallback behavior. Current English confirmation support is more extensive. 6. Decide the deployment model: USB-connected PC ASR/CCBM is development mode. Independent headset deployment needs reachable

Validation still required

services or another runtime. 7. Validate any desired real mannequin measurements via sensors/BLE. Do not label hand excursion or spoken breaths as verified physical measurements. 8. Clean up inactive AED/QCPR branches only after reviewing the intended future scope, then produce a reviewable commit/PR. Local changes are not automatically incorporated into Denis's GitHub repository.

Marek 1/14 • marek_offer

German: Ich bin hier. Was soll ich tun?

Polish: Jestem tutaj. Co mam robić?

English translation: I’m here. What should I do?

When: After accepted responsiveness assessment, with a 2.5-second delay.

Marek 2/14 • marek_call

German: Ich rufe die 112 an und stelle auf Lautsprecher.

Polish: Dzwonię pod 112 i włączam głośnik.

English translation: I’m calling 112 and putting it on speakerphone.

When: Accepted emergency-call action.

Marek 3/14 • marek_dispatch

German: Die Leitstelle ist dran. Sie fragen, ob er normal atmet.

Polish: Dyspozytor jest na linii. Pyta, czy on normalnie oddycha.

English translation: The dispatcher is on the line. They’re asking whether he is breathing normally.

When: Five seconds after the emergency-call action.

Marek 4/14 • marek_prompt_call

German: Soll ich den Notruf wählen?

Polish: Mam zadzwonić na numer alarmowy?

English translation: Should I call the emergency number?

When: CALL_HELP: no call after 15 seconds.

Marek 5/14 • marek_go_aed

German: Ich hole den Defibrillator!

Polish: Idę po defibrylator!

English translation: I’ll get the defibrillator!

When: Accepted AED request.

Marek 6/14 • marek_prompt_aed

German: Soll ich einen Defibrillator suchen?

Polish: Mam poszukać defibrylatora?

English translation: Should I look for a defibrillator?

When: During CPR, call made but no AED requested: more than 20 seconds since the call.

Marek 7/14 • marek_back_aed

German: Hier ist der AED! Machen Sie ihn an.

Polish: Tu jest AED! Proszę go włączyć.

English translation: Here is the AED! Turn it on.

When: Return with AED after the fetch sequence; travel affects timing.

Marek 8/14 • marek_faster

German: Die Leitstelle sagt: schneller drücken!

Polish: Dyspozytor mówi: szybciej uciskać!

English translation: The dispatcher says: press faster!

When: Four consecutive rate samples below 98 BPM after warm-up; call/phone available; cooldown.

Marek 9/14 • marek_slower

German: Etwas langsamer, sagt die Leitstelle.

Polish: Trochę wolniej, mówi dyspozytor.

English translation: A little slower, says the dispatcher.

When: Four consecutive rate samples above 124 BPM after warm-up; call/phone available; cooldown.

Marek 10/14 • marek_position

German: Die Hände in die Mitte der Brust, sagt die Leitstelle.

Polish: Ręce na środek klatki piersiowej, mówi dyspozytor.

English translation: Hands in the centre of the chest, says the dispatcher.

When: Three consecutive off-target strokes; call/phone available; cooldown.

Marek 11/14 • marek_resume

German: Das Gerät sagt: sofort weiterdrücken!

Polish: Urządzenie mówi: natychmiast uciskać dalej!

English translation: The device says: resume compressions immediately!

When: Post-shock inactivity longer than five seconds.

Marek 12/14 • marek_swap

German: Ich kann beim Drücken übernehmen, wenn Sie nicht mehr können.

Polish: Mogę zmienić przy uciskaniu, jeśli trzeba.

English translation: I can take over compressions if you can’t continue.

When: Analysis-due event during CPR; this line offers help rather than accepting a swap.

Marek 13/14 • marek_ems

German: Der Rettungsdienst ist da.

Polish: Karetka już jest.

English translation: The ambulance crew is here.

When: Handover, or eight seconds after simulated recovery/ROSC.

Marek 14/14 • marek_take_over

German: Ich drücke weiter – kleben Sie die Elektroden!

Polish: Ja uciskam dalej – proszę naklejać elektrody!

English translation: I’ll keep doing compressions—attach the pads!

When: AED handoff/helper-compression sequence.

Runtime file inventory

scripts/ai/audio_manager.gd • Changed • 386

scripts/ai/language_manager.gd • Changed • 260

scripts/ai/llm_orchestrator.gd • Changed • 142

scripts/ai/observation_interpreter.gd • Added • 76

scripts/api/uni_api_client.gd • Changed • 219

scripts/ekg/ekg_system.gd • Unchanged • 71

scripts/feedback/score_tracker.gd • Unchanged • 59

scripts/feedback/session_logger.gd • Unchanged • 64

scripts/main.gd • Changed • 856

Runtime file inventory

scripts/mannequin/aed_device.gd • Unchanged • 547

scripts/mannequin/breathing_system.gd • Unchanged • 44

scripts/mannequin/bystander_controller.gd • Unchanged • 218

scripts/mannequin/heart_system.gd • Unchanged • 68

scripts/mannequin/mannequin_controller.gd • Changed • 312

scripts/mannequin/pupil_system.gd • Unchanged • 23

scripts/mannequin/ragdoll.gd • Unchanged • 383

scripts/mannequin/rig_animator.gd • Unchanged • 379

scripts/qcpr/qcpr_backend.gd • Unchanged • 69

Runtime file inventory

scripts/qcpr/qcpr_ble_real_backend.gd • Unchanged • 168

scripts/qcpr/qcpr_constants.gd • Unchanged • 179

scripts/qcpr/qcpr_manager.gd • Unchanged • 195

scripts/qcpr/qcpr_packet.gd • Unchanged • 259

scripts/qcpr/simulated_manikin_backend.gd • Unchanged • 478

scripts/scenario/bystander_director.gd • Unchanged • 498

scripts/scenario/ccbm_bridge.gd • Added • 120

scripts/scenario/scenario_controller.gd • Changed • 371

scripts/scenario/single_witness_director.gd • Added • 25

Runtime file inventory

scripts/scenario/speech_confirmation.gd • Added • 48

scripts/scenario/spoken_observation.gd • Added • 70

scripts/tracking/aruco_tracker.gd • Changed • 207

scripts/tracking/colored_camera.gdshader • Added • 28

scripts/tracking/colored_marker_image.gd • Added • 27

scripts/tracking/head_tilt_gesture.gd • Added • 90

scripts/tracking/marker_layout.gd • Added • 104

scripts/tracking/marker_pose_gizmo.gd • Added • 42

scripts/tracking/openxr_setup.gd • Changed • 70

Runtime file inventory

scripts/tracking/room_anchor.gd • Unchanged • 7

scripts/tracking/room_layout.gd • Changed • 178

scripts/tracking/spatial_anchor.gd • Changed • 357

scripts/tracking/startup_placement.gd • Added • 99

scripts/tracking/user_camera_provider.gd • Added • 434

scripts/tracking/viewer_motion_gate.gd • Added • 156

scripts/tracking/xr_interaction.gd • Changed • 466

scripts/ui/briefing_panel.gd • Changed • 149

scripts/ui/calibration_panel.gd • Changed • 219

Runtime file inventory

scripts/ui/dashboard.gd • Changed • 364

scripts/ui/ekg_renderer.gd • Unchanged • 31

scripts/ui/hand_menu.gd • Changed • 773

scripts/ui/info_panel.gd • Changed • 111

scripts/ui/qcpr_debug_panel.gd • Unchanged • 217

scripts/utils/performance_monitor.gd • Unchanged • 34

What to test before calling the app finished

Placement: stationary mannequin, head motion, marker loss, restart and recenter; verify saved nudges and floor height.

Exercise: full English run with natural multi-sentence speech, interruptions, 30 compressions, two reported breaths and handover; then German/Polish.

Deployment: PC/USB ASR and CCBM remain development dependencies. Connect real mannequin sensors if verified depth/recoil/ventilation are required.

Sources and scope of this review

Denis baseline: NaReT-Nachbarsprache-wird-reanimiert/naret-xr-app, commit 9d880a1 (28 September 2026).

Current: local working tree, installed version 42 / 0.3.40-original-stabilizer. Paul addon: 729f8c0.

Full source audit: DENIS_FINAL_APP_REVIEW.md. Reported validation: 24 application suites passed with the blocked Windows native detector excluded. Hardware behavior still needs the checks listed here.